Deep Learning in Financial Markets: A Systematic Literature Review of Methods and Future Direction for Price Prediction

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***Abstract*—This systematic literature review (SLR) investigates the application of deep learning methods in financial market price prediction, focusing on the most studied financial markets, commonly used deep learning techniques, input variables, performance evaluation metrics, and proposed future research directions. The review covers 56 peer-reviewed articles published between 2020 and 2024, selected from ACM Digital Library, IEEE Xplore, and ScienceDirect databases, examining the stock, cryptocurrency, commodity, and forex market price prediction. Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) are identified as the most widely used methods due to their capability in handling long-term dependencies in time series data. The review highlights future directions in hyperparameter optimization, hybrid model integration, and the use of social network data. This study fills the gap in literature by analyzing recent advancements and their applicability across various financial markets.**

Keywords—systematic literature review, deep learning, financial market, price prediction, stock, commodity, forex, cryptocurrency, LSTM, GRU, hyperparameter optimization, market sentiment, hybrid models.

I. INTRODUCTION

In recent years, price prediction in financial markets has become an increasingly popular and rapidly developing research topic. Nevertheless, the non-linear, complex, and volatile nature of financial markets, along with the numerous factors that influence price movements—such as historical prices, trading volumes, political events, news, company financial reports, market sentiment, and macroeconomic data—pose significant challenges for accurate price forecasting [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23].

These challenges have become a key focus in the field of financial market price prediction. Various approaches have been developed to address these issues. One of the most widely used methods today is the development of deep learning-based models [6], [7], [10], [12], [15], [16], [17], [18], [21], [24], [25], [26], [27], [28], [29]. A common strategy involves building hybrid models, either by combining different deep learning models or integrating deep learning with conventional or statistical models. In addition, model development often incorporates optimization techniques, decomposition, and attention mechanisms. Multivariate and multistep models are also increasingly being explored. The factors used in financial market price prediction models are becoming more diverse. In addition to historical data such as OHLC (Open, High, Low, Close) prices and trading volumes, recent models also take into account technical indicators, macroeconomic data, and market sentiment derived from social media as important variables in predicting price movements.

This Systematic Literature Review (SLR) aims to provide a comprehensive analysis of the application of deep learning techniques in financial markets. Previous reviews have not fully explored the diverse financial markets, especially the role of emerging markets like cryptocurrency and commodities. This paper aims to bridge this gap by providing a comprehensive review of deep learning applications in price prediction across stock, forex, cryptocurrency, and commodity markets. The primary contribution of this study is the identification of current trends, common techniques, and the exploration of emerging hybrid models. Moreover, it highlights gaps in previous research, particularly in integrating social sentiment and macroeconomic factors into prediction models. By examining a wide range of research studies, the review seeks to identify which financial markets have received the most attention in deep learning research, determine the most commonly used methods and variables for price prediction, and establish the metrics most frequently employed to evaluate the performance of deep learning-based models. Additionally, this review aims to propose future research directions in this field. Therefore, the research questions explored in our systematic literature review are outlined.

- RQ1: Which financial markets types have the most attention in Deep Learning research?
- RQ2: Which Deep Learning methods are most used for price predictions in Financial Market?
- RQ3: What variables are commonly utilized for deep learning-based price predictions in financial markets?
- RQ4: Which metrics are most commonly used to verify the performance of Deep Learning based price prediction models in Financial Markets?

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- RQ5: What are the proposed future works in the explored studies?

II. Methodology

A. Search Strategy

Publication searches were conducted in several publication databases, including: ACM Digital Library, IEEE Xplore & Access, and Science Direct. The keywords used for searching in the ACM Digital Library and ScienceDirect are: (stock OR forex OR gold OR XAU OR crypto OR "financial market") AND (predict OR forecasting) AND "deep learning". Meanwhile, the keywords used for searches on IEEE Xplore & Access are: 1)(stock OR forex OR gold OR XAU OR crypto OR "financial market") AND (predict OR forecasting) AND "deep learning". 2) (study OR literature OR review) AND (stock OR forex OR "financial market") AND (predict OR forecasting) AND "deep learning".

B. Selection Criteria

The article selection process is carried out in several stages to filter a collection of articles from various databases. The initial search found 576 related publications, furthermore we filtered tittle, abstracts, and inclusion/exclusion criteria. To ensure the relevance of the topic and the recency of the articles, we limited the publication years to between 2020 and 2024 then obtained 136 articles. After carrying out a review process, 56 articles were finally used in this paper.

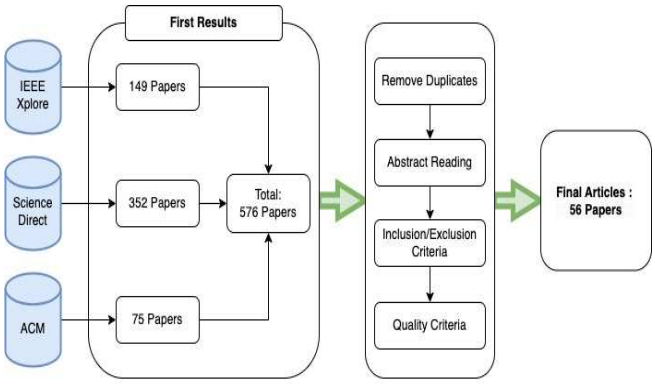


Fig. 1. Literature search and selection

C. Data Extraction and Management

In conducting a SLR, the data extraction process is an important step to ensure that the information collected is relevant and reliable. Each selected document will be reviewed and relevant data will be extracted. The extracted data includes publication year, author, title, deep learning model used, analyzed dataset, evaluation metrics, model performance results, and future research proposals. To manage the extracted data, we use reference management software or spreadsheets.

D. Published Report

The report consists of an introduction, methodology, results, discussion, and conclusion. The results are displayed using tables and charts to enhance clarity, and the document has been submitted to an appropriate journal for publication.

III. Result

A. Financial Market Types in Deep Learning Research

There are four main types of financial markets, which include stock markets, foreign exchange markets, commodity markets, and cryptocurrency exchange markets [30]. Stocks are the most researched market [1], [2], [3], [4], [5], [6], [7], [8], [11], [12], [16], [17], [18], [19], [21], [22], [23], [24], [26], [27], [28], [29], [30], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], followed by the increasingly popular cryptocurrency market [13], [20], [25], [33], [34], [35], [36], then commodities like gold and oil [9], [10], [15], [32], and finally the Forex market, which includes major currency pairs [14], [31] as depicted in Figure 2.

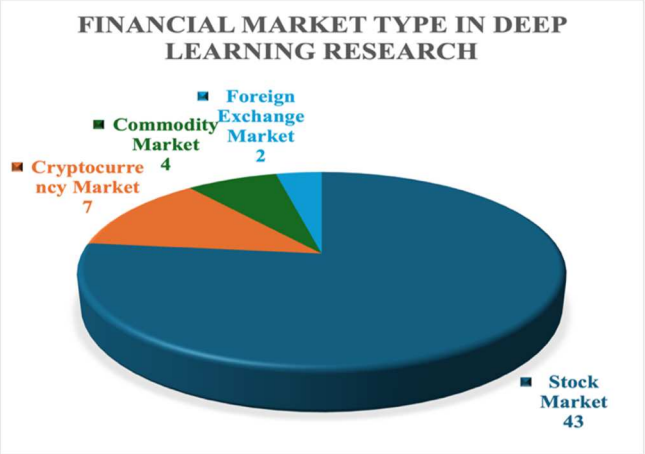


Fig. 2. Financial market type in deep learning research

B. Deep Learning Methods in Financial Market Prediction

The most widely used deep learning method for predicting prices in financial markets is Long Short-Term Memory (LSTM), followed by Gated Recurrent Units (GRU)-based models [3], [4], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [32], [33], [34], [37], [38], [42], [43], [44], [45], [46], [48], [50], [51], [52], [53], [54]. Both excel at handling long-term dependencies in time series data. Convolutional Neural Network (CNN), Transformer, and Generative Adversarial Network (GAN)-based methods are also commonly found in the reviewed research [1], [2], [5], [15], [16], [17], [18], [20], [26], [27], [28], [39], [40], [41], [48], [49]. Other methods, such as Deep Q-Network (DQN), Deep Neural Network (DNN), Multilayer Perceptron (MLP), and Temporal Convolutional Network (TCN), appear in several studies but are less frequently used [7], [12], [17], [48], [50], [55].

Additionally, various techniques aimed at improving accuracy are often employed, such as optimization techniques, decomposition, and Attention Mechanisms [4], [6], [7], [9], [11], [16], [17], [19], [24], [26], [27], [34], [42], [52]. Optimization techniques like Particle Swarm Optimization (PSO) are mentioned in several studies as well. Some research has developed models based on multivariate or multistep approaches [3], [25], [27], [54]. To further enhance accuracy, hybrid deep learning models have been developed in some studies, integrating different deep learning models, statistical models, or other techniques previously mentioned [6], [7], [10], [12], [15], [16], [17], [18], [21], [24], [25], [26], [27], [28], [29].

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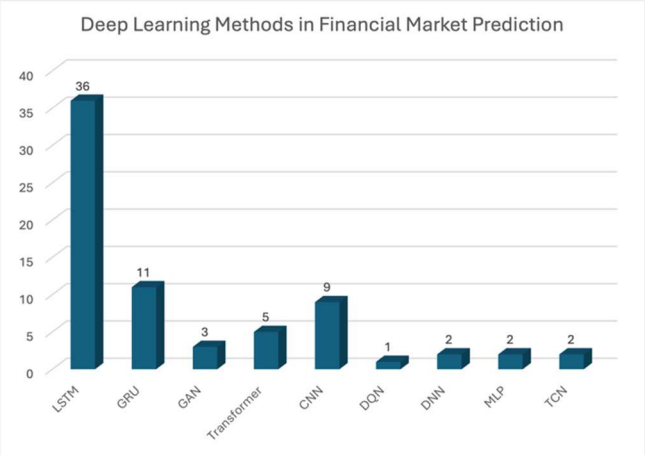


Fig. 3. Deep learning methods in financial market prediction

C. Variables in Financial Market Predictions

The variables used for deep learning-based predictions in financial markets can be categorized into four main concepts. First, market prices, including OHLC prices, adjusted close prices, volume, and turnover, provide fundamental data for analyzing price movements and market activity [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [24], [25], [27], [28], [29], [31], [32], [33], [34], [35], [37], [38], [40], [41], [42], [43], [44], [45], [46], [47], [48], [50], [51], [52], [53]. Second, macroeconomic, fundamental, and external factors are also included in prediction models, such as currency exchange rates, inflation, and interest rates, as well as major market indices like the DJIA, S&P 500, and Nasdaq [7], [9], [10], [25], [28], [33], [34], [51].

Third, technical indicators such as Simple Moving Average (SMA), Exponential Moving Average (EMA), Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI) and Average Directional Index (ADX) help capture trend patterns and momentum in market prices [4], [7], [11], [20], [21], [28], [46], [51]. Finally, market sentiment and social data, including Twitter texts, Google search trends, and financial news sentiment, are leveraged to assess public opinion and predict market behavior [25], [28], [33], [35], [40]. Additionally, for cryptocurrencies, specific data such as on-chain data, hash rate, and active transaction numbers are used to predict the movement of digital assets [26], [34], [35]. These variables collectively support accurate deep learning-based predictions in various financial markets.

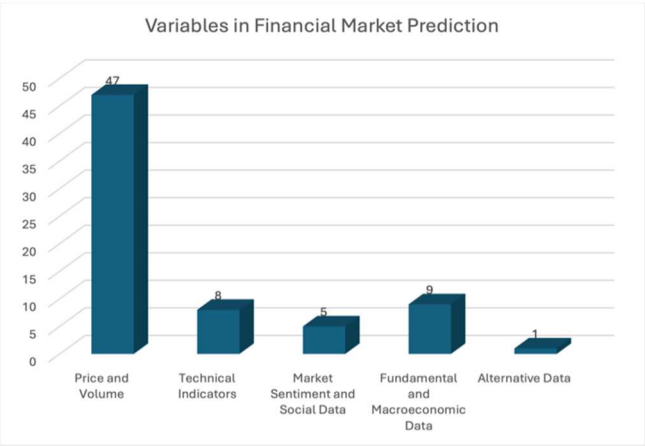


Fig. 4. Variables in financial market prediction

D. Performance Metrics in Financial Market Predictions

The most commonly used metrics to verify the performance of Deep Learning-based price prediction models in financial markets can be categorized into three main groups: Error Metrics, Statistical & Correlation Metrics, and Classification Metrics. Among the Error Metrics, RMSE (Root Mean Square Error), MAE (Mean Absolute Error), MSE (Mean Squared Error), and MAPE (Mean Absolute Percentage Error) are frequently used to measure the deviation of predictions from actual values, with RMSE being the most widely referenced. In the Statistical & Correlation Metrics group, R^2 (R-squared) is the most popular for assessing how well the model explains data variability, while Theil's U and Explained Variance Score (EVS) are also used for model comparison and variability measurement. Although less common, Classification Metrics such as Accuracy, Precision, Recall, and F1 Score are sometimes applied to evaluate prediction accuracy, particularly when classification tasks are involved. Overall, RMSE, MAE, MSE, MAPE, and R^2 are the most frequently used metrics for verifying the performance of financial price prediction models.

TABLE I. PERFORMANCE METRICS IN FINANCIAL MARKET PREDICTIONS

No	Metric	Description	Citation
1	Error Metrics	Measures the magnitude of prediction errors compared to actual data. Common metrics include: RMSE, MAE, MSE, and MAPE.	[1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [16], [17], [18], [19], [21], [22], [24], [25], [26], [27], [28], [29], [31], [32], [33], [34], [37], [38], [39], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [54]
2	Statistical & Correlation Metrics	Measures statistical correlation and relationships between model predictions and actual data.	[4], [5], [6], [7], [9], [11], [12], [13], [19], [20], [24], [26], [28], [29], [33], [34], [37], [38], [42], [43], [44], [45], [48], [50], [51]
3	Classi fication Metrics	Used to evaluate model performance in classification tasks.	[3], [23], [30], [35], [40], [44], [56]

E. The Proposed Future Works in The Explored Studies

Future research proposals related to price prediction in Financial Markets encompass several key directions. First, hyperparameter optimization such as learning rate, number of epochs, batch size, and time series window length using evolutionary techniques like PSO and Genetic Algorithm (GA) is a focus to improve the performance of deep learning models. Additionally, other metaheuristic approaches can be applied to improve the performance of deep learning models in predicting financial market prices [4], [6], [7], [22], [23], [24], [25], [28], [34], [38], [46], [49], [50], [51], [52], [54], [55]. Second, combining multiple models, such as bagging, boosting, and stacking, can be applied to enhance predictive accuracy. Moreover, integrating deep learning methods with traditional models like LSTM, GRU, or CNN, along with evolutionary optimization techniques, can improve model performance by handling complex and non-linear data in financial markets [3], [7], [10], [13], [17], [20], [24], [28],

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[34], [35], [36], [49], [54]. Third, utilizing social networks, investor sentiment, and Natural Language Processing (NLP) techniques can enhance model accuracy by understanding the context and semantics of textual data. Sentiment analysis using NLP will also play a key role, where text data from social media, news, and comments can be used to capture investor sentiment, thereby improving model accuracy in predicting market volatility [7], [9], [13], [17], [30], [32], [33], [34], [42], [51].

Additionally, multivariate time series modelling with various time scales and the integration of macro and microeconomic factors is proposed to enrich the information used in prediction models [11], [18], [19], [34], [48]. The use of technical indicators and macroeconomic factors, such as inflation and oil prices, should also be expanded to more accurately predict various financial assets [9], [15], [21], [26], [34], [35], [40], [41], [48], [53], [56]. Lastly, applying prediction models across different financial products, such as stocks, cryptocurrency, commodities, and futures, presents opportunities to test the model's effectiveness in various asset types and explore cross-market correlations that influence volatility and price predictions [6], [17], [21], [24], [26], [27], [34]. Thus, future research directions in financial market price prediction involve a multidisciplinary approach that includes hyperparameter optimization, hybrid models, social networks, multivariate time series forecasting, external macroeconomic factors, and applications across various financial products as depicted in Figure 5.

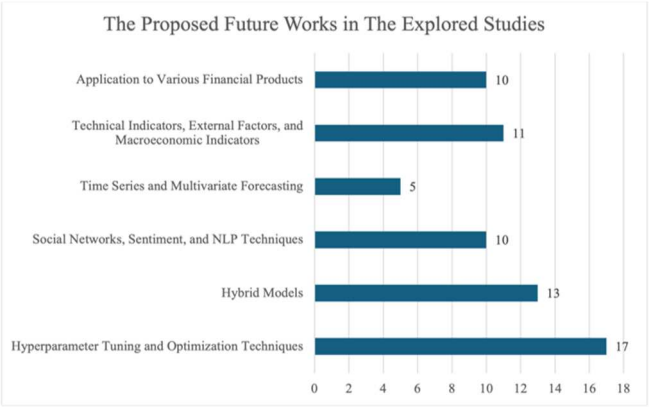


Fig. 5. The proposed future works in the explored studies

IV. Discussion

A. Deep Learning Methods

In our investigation of deep learning methods, we found Recurrent Neural Network (RNN) methods and their variants such as LSTM and GRU are the most widely used deep learning methods for predicting prices in financial markets. LSTM and GRU excel at handling long-term dependencies in time series data [8], [29], [42]. Additionally, various techniques aimed at improving accuracy are often employed, such as optimization techniques, decomposition, and Attention Mechanisms [4], [6], [7], [9], [11], [16], [19], [24], [26], [27], [34], [42], [52].

Deep learning models can occasionally be overfit to the training data. It is essential to prevent both overfitting and underfitting to ensure accurate long-term predictions [54]. To effectively address the problems of overfitting, underfitting, and vanishing/exploding gradients in LSTM neural networks, PSO can be applied. By utilizing PSO, the optimal values for the initial input weights, recurrent weights, and biases in the LSTM network can be determined, leading to improved model performance and stability [38]. Hybrid Deep Learning models designed to tackle the complexities of financial market prediction, including nonlinearity, market volatility and intricate patterns [23] [30] [32] [43]. To further enhance accuracy, hybrid deep learning models have been developed in some studies, integrating different deep learning models, statistical models, or other techniques [6], [7], [10], [12], [15], [16], [17], [18], [21], [24], [25], [26], [27], [28], [29].

B. Future Direction

The results of this SLR indicate several promising directions for future research in financial market price prediction. These future directions can be broadly categorized into hyperparameter optimization, model integration, sentiment analysis, multivariate time series modeling, and cross-market applications. Each of these areas presents unique challenges and opportunities for advancing predictive accuracy and reliability in financial markets.

One of the most recurring themes in the literature is the need for hyperparameter optimization to enhance deep learning model performance. Techniques such as PSO and GA have been proposed as efficient approaches for tuning hyperparameters such as learning rate, number of epochs, batch size, and time series window length [4], [6], [7], [22], [23], [24], [25], [28], [34], [38], [46], [49], [50], [51], [52], [54], [55]. These evolutionary techniques, along with other metaheuristic methods, offer potential improvements in the performance of deep learning models by finding optimal configurations that may not be easily attainable through manual tuning or traditional methods on experimenting with these optimization techniques across various models and financial market datasets to better understand their impact on prediction accuracy.

Another critical direction for future research is the combination of multiple models. Techniques such as bagging, boosting, and stacking have been highlighted as methods that can enhance predictive accuracy by aggregating the strengths of different models. Integrating deep learning models like LSTM, GRU, or CNN with traditional statistical models and evolutionary optimization techniques is proposed to tackle complex and nonlinear financial market data. This hybrid approach could provide more robust predictions by capturing diverse patterns in the data [3], [7], [10], [13], [17], [20], [24], [28], [34], [35], [36], [49], [54]. Incorporating sentiment analysis using NLP techniques represents another area of future research [7], [9], [13], [17], [30], [32], [33], [34], [42], [51]. By analyzing textual data from social networks, news articles, and investor comments, researchers can capture investor sentiment, which can serve as a valuable predictor of market movements. Sentiment analysis can help improve the accuracy of models by providing insights into market sentiment and volatility. Future studies could delve deeper into understanding how social media, news sentiment, and other textual data sources influence financial markets, especially in fast-moving markets like cryptocurrency.

The future of financial market price prediction research will likely involve a multidisciplinary approach that includes advances in hyperparameter optimization, hybrid models, sentiment analysis, multivariate time series forecasting, and cross-market applications. As financial markets become increasingly complex and interconnected, these directions hold promise for improving predictive accuracy and fostering

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a deeper understanding of the factors that drive market behavior.

V. Conclusion

This systematic literature review highlights the growing application of deep learning techniques in financial market price prediction, identifying key methods, variables, and future research directions. Financial markets, such as stocks, cryptocurrencies, commodities, and forex, have been at the forefront of deep learning research, with stocks being the most extensively studied market. Among the deep learning methods, LSTM and GRU are the most frequently employed, thanks to their ability to handle long-term dependencies in time series data. Other methods such as CNN, Transformer models, and GAN have also gained traction in recent studies. The variables commonly used in deep learning-based price predictions range from historical market prices to technical indicators, market sentiment, and macroeconomic factors. Sentiment analysis, driven by NLP techniques, has shown promise in improving model performance by incorporating textual data from social networks, financial news, and investor comments. Performance metrics such as RMSE, MAE, MSE, and R^2 are the most popular for evaluating the accuracy of price prediction models. These metrics help quantify the model's prediction errors and assess its ability to explain the variability in financial data.

This review contributes to the literature by providing a detailed comparison of deep learning methods used in financial market prediction. It highlights the growing role of hybrid models and the increasing importance of incorporating sentiment analysis into prediction frameworks. Future research directions indicate the need for more advanced hyperparameter optimization techniques like PSO and GA. The integration of multiple models, hybrid deep learning approaches, and multivariate time series modelling holds significant potential for improving predictive accuracy. Additionally, expanding the use of social network data, macroeconomic factors, and testing models across various financial products will enhance the robustness and applicability of future predictive models.

In conclusion, deep learning remains a promising area of research for financial market price prediction, offering new avenues for improvement through optimization, hybrid models, and the incorporation of diverse data sources. Future studies should aim to refine these methods and expand their application to gain a deeper understanding of the dynamics that drive market prices.

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